1. Behavior

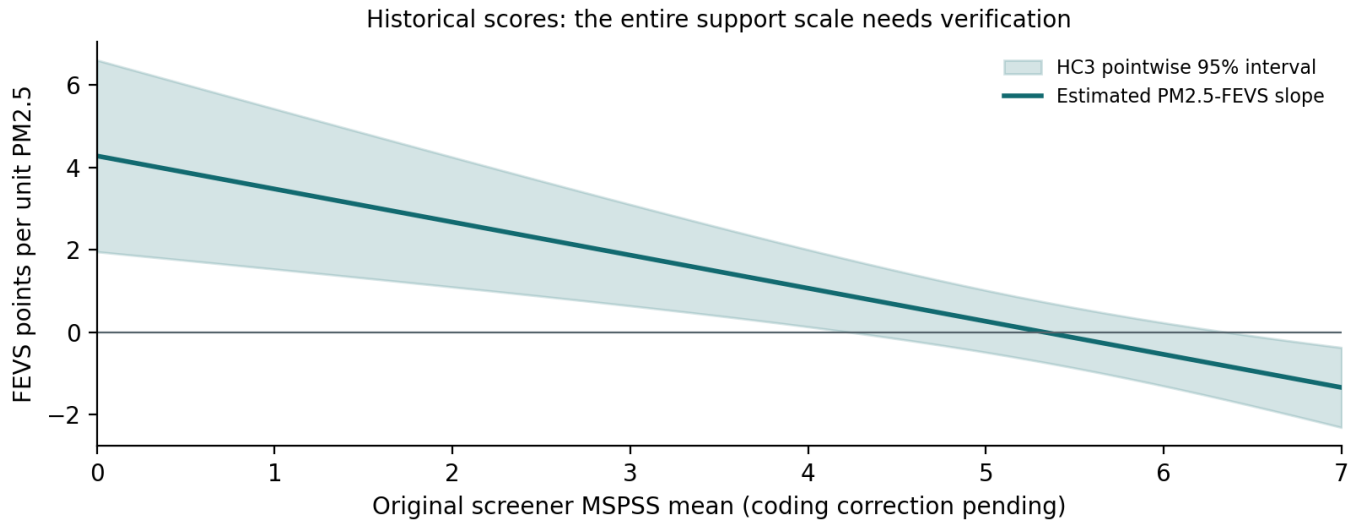
Does perceived support modify the PM2.5-FEVS association?

PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

FEVS is the primary financial-vulnerability outcome. These age/gender-adjusted models ask whether the PM2.5 association varies with perceived support. The displayed findings use the original screener MSPSS scores and remain provisional pending their coding correction.

Content



Historical fit	Interaction b	HC3 p (uncorrected sensitivity check)
baseline	-0.802	0.000168
adi	-0.802	0.000224
ecog	-0.714	0.001093
adi+ecog	-0.711	0.001313

The displayed conditional-slope plot uses the old scores. Pointwise intervals and any JN regions are exploratory, not simultaneous bands. Original primary behavioral interaction: b=-.802, HC3 CI [-1.208,-.396], p=.000168; Holm p=.000336 across the original two primary tests. Those numbers are not scoring-corrected findings.

Conclusion

In the original export, the PM2.5-FEVS association becomes less positive as reported support increases, and the interaction persists after adding ADI/Ecog. That is the candidate behavioral pattern, not a finalized buffering result: the coding correction can affect all 97 scores, not just five low values. Verify both MSPSS administrations, assess agreement, then rerun the affected models and original multiplicity summaries without changing the cohort.

The focal FEVS neural models shown earlier do not include MSPSS and are unaffected by its correction. No matching corrected neural PM2.5 x support result has been established. Separate behavioral and neural associations do not demonstrate mediation.

1. Behavior

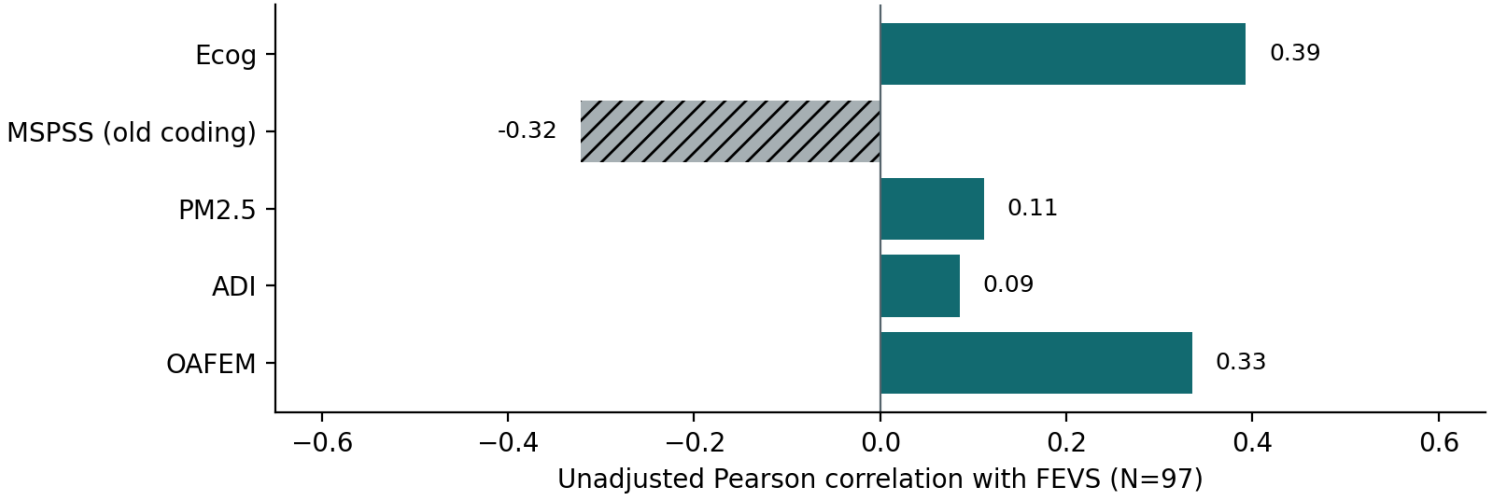
How does financial vulnerability relate to the other measures?

PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Before linking behavior to brain responses, inspect the already-computed marginal associations. These descriptive correlations are not adjusted effects, causal paths or criteria for choosing whichever model is significant.

Content



Measure paired with FEVS	Pearson r	Spearman rho	N
OAFEM (secondary)	0.33	0.35	97
ADI	0.09	0.08	97
PM2.5	0.11	0.12	97
MSPSS (old coding; pending)	-0.32	-0.36	97
Ecog	0.39	0.30	97

Conclusion

FEVS has modest descriptive associations with Ecog and OAFEM, but weak marginal associations with ADI and PM2.5. The MSPSS association needs rescoring. Neither weak exposure correlations nor the support interaction establish an environment-brain-FEVS pathway.

Next specific question: does Ecog agree with objective cognitive performance? Inventory NIH EXAMINER/TabCAT executive measures, Salthouse Letter/Pattern Comparisons, and WIAT-IV Math Fluency, with assessment dates and availability within N=97. No comparison results are available yet; no cohort or neural-model changes are implied.

2. Basic Activation

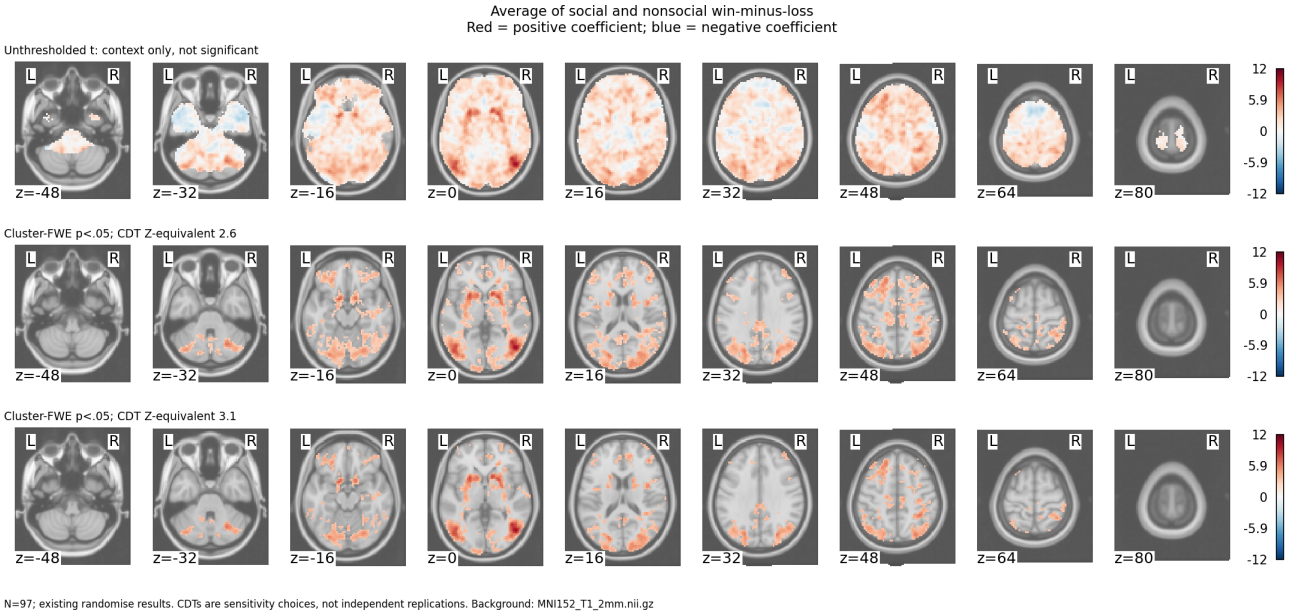
Do wins elicit the expected striatal response?

PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Yes: both Social Doors and nonsocial Doors show bilateral ventral-striatal-region win > loss responses. This is expected-effect/task validation, not a preregistered confirmatory analysis. Social specificity is a separate question.

Content



Win > loss outcome	Left / right regional peak t	Left / right cluster-FWE p	Anatomical VS mean [descriptive 95% CI]
Social Doors	7.56 / 8.96	0.0007 / 0.0001	60.98 [45.02, 76.94]
Nonsocial Doors	9.85 / 10.78	0.0003 / 0.0003	72.28 [55.77, 88.78]
Across-task average	11.79 / 11.70	0.0001 / 0.0001	66.63 [53.24, 80.02]

Conclusion

The expected striatal reward response is present, not missing: averaged win > loss peaks near MNI [-12.7, 6.3, -9.6] and [14.3, 9.0, -9.6] lie in corrected clusters with FWE p=.0001. Both tasks therefore recover the expected reward response; this is task validation, not the novel financial-vulnerability result.

Figure rows: unthresholded context, corrected CDT 2.6, corrected CDT 3.1. Red/blue show positive/negative coefficients. Cluster p-values are not voxel-FWE claims. ROI summaries are descriptive COPE units, not percent signal change or new ROI tests. Existing no-TEDANA sm-5 FEAT inputs and exclusions are unchanged.

2. Basic Activation

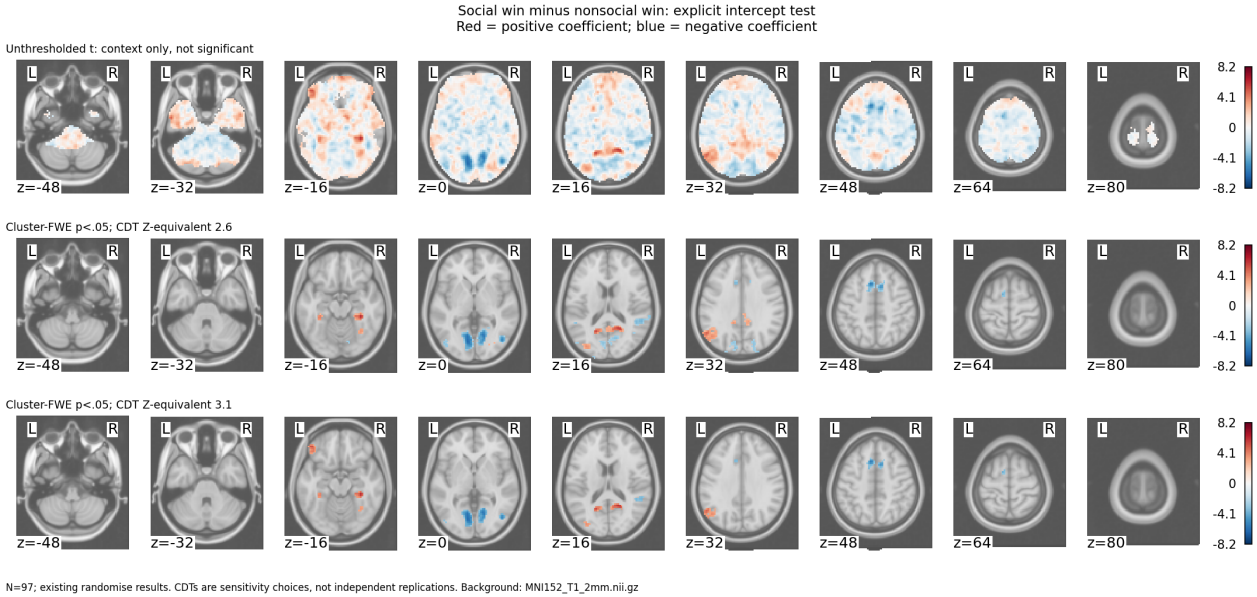
Do winning responses differ by social context?

PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Compare social win minus nonsocial win. This is distinct from the double subtraction, which additionally references losses.

Content



CDT (Z-equivalent)	Direction	Minimum cluster-FWE p	Corrected voxels
2.6	positive	0.0175	1196
2.6	negative	0.0019	1991
3.1	positive	0.0021	762
3.1	negative	0.0011	1008

Conclusion

There are regions with social > nonsocial winning responses and regions showing the reverse, at both thresholds. This establishes domain differentiation for winning, not a universal social-reward increase or a domain-by-valence interaction. It should not be presented as proof of additional striatal activation for social wins.

The displayed domain effects include temporal and posterior cortical areas, with both positive and negative regions. These are coarse visual descriptions, not new atlas assignments; exact striatal overlap for this contrast has not been quantified.

Both new intercept jobs and four directional maps passed validation, including the native sign-flip gate; 10,000 permutations, residual df=90. The pre-existing win-loss and double-subtraction tests were reused, not rerun.

2. Basic Activation

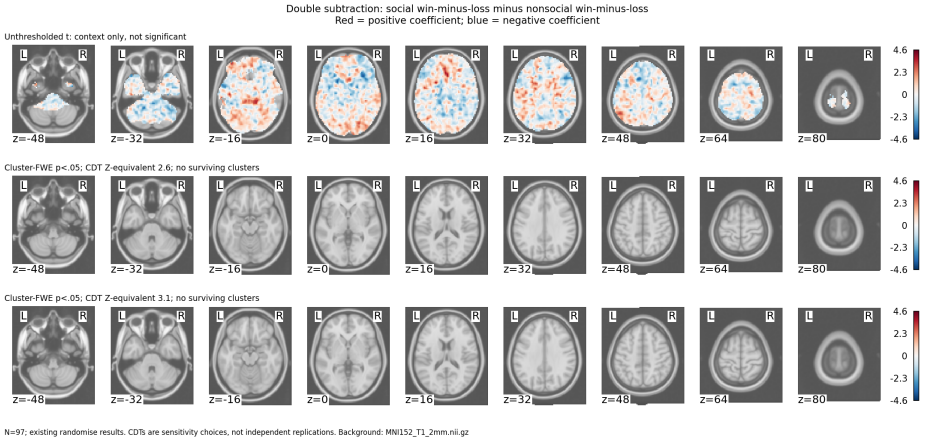
Is win-loss differentiation stronger for social outcomes?

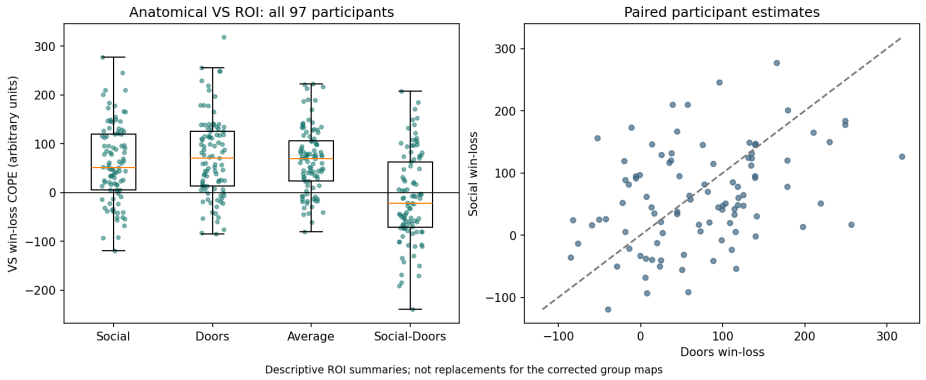
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Test (social win-loss) minus (nonsocial win-loss). A social-winning difference alone does not answer this domain-by-valence question.

Content





CDT	Direction	Minimum cluster-FWE p	Corrected voxels
2.6	positive	0.4232	0
2.6	negative	0.2905	0
3.1	positive	0.2414	0
3.1	negative	0.2081	0

ROI values are arbitrary COPE units, not percent signal change. The anatomical seed-VS mask was intersected with the common analysis mask, and all 97 participants remain. These are descriptive ROI summaries, not new small-volume-corrected tests. The ROI intervals do not replace the whole-brain cluster-corrected tests.

Conclusion

No corrected group-average domain-by-valence effect was detected at either CDT. The anatomical VS social-minus-nonsocial win-loss mean is -11.29 [descriptive 95% CI -29.65, 7.06]. This does not establish equivalence and does not rule out individual-difference associations. The contrast is not social win minus nonsocial win, because it also subtracts losses.

3. Exploratory Activation

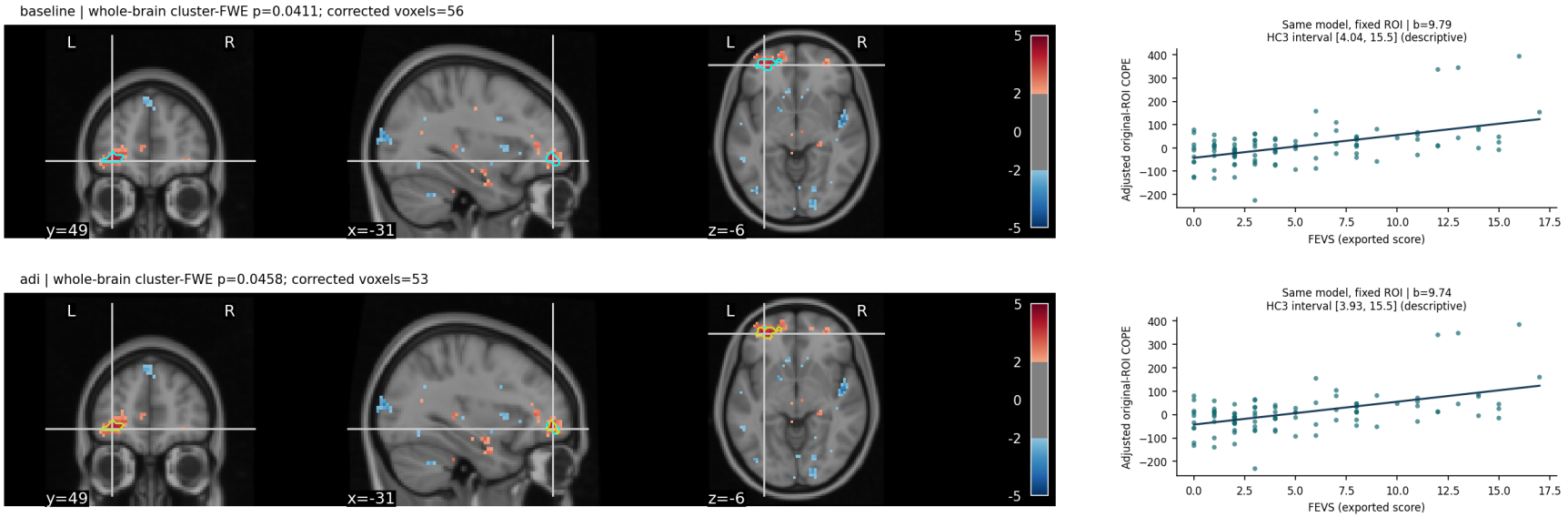
Does FEVS relate to social-winning activation?

PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Does financial vulnerability relate to the social win-minus-nonsocial win activation contrast? Rows: original and +ADI.

Content



Each brain row is beside its exact model-adjusted scatter. Cyan: original corrected cluster; gold: adjusted corrected cluster. The same original ROI is used for every scatter, including whole-brain null fits. Brain shading |t|>=2 is descriptive, not corrected significance.

Model	Cluster-FWE p, CDT 3.1	ROI slope / HC3 interval (descriptive)	Leave-one-out slope range / max Cook's D
baseline	0.0411	9.79 [4.04, 15.5]	[7.86, 10.4] / 0.206
adi	0.0458	9.74 [3.93, 15.5]	[7.85, 10.4] / 0.294

Conclusion

The original association is positive, but adding Ecog removes whole-brain corrected significance. Treat this as qualified evidence, not an adjustment-robust cortical vmPFC/OFC finding.

Anatomy caveat: original clusters are predominantly white-matter-labeled in a standard atlas; the jointly adjusted conditional cluster is approximately 64% white-matter-labeled. The conditional activation centroid shifts only 1.1 mm with joint adjustment. Neither fact validates a clean cortical vmPFC/OFC claim.

All 97 participants retained. Selected-ROI scatter slopes/HC3 intervals are descriptive only, not independent confirmation or a rescue of a null whole-brain result. No cases removed; leave-one-out checks do not exclude collective influence. Findings pass only at CDT 3.1; correction is per map, not across the exploratory model family. None of these focal models contains MSPSS.

3. Exploratory Activation

Does the activation association persist with Ecog adjustment?

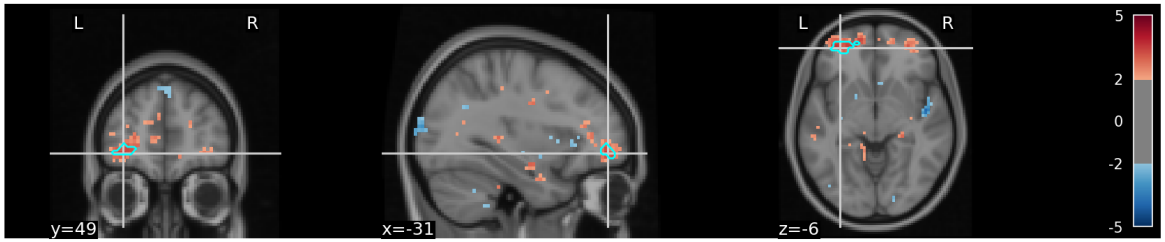
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

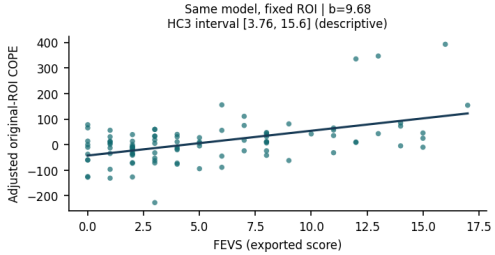
Context

Does financial vulnerability relate to the social win-minus-nonsocial win activation contrast? Rows: +Ecog and +ADI/Ecog.

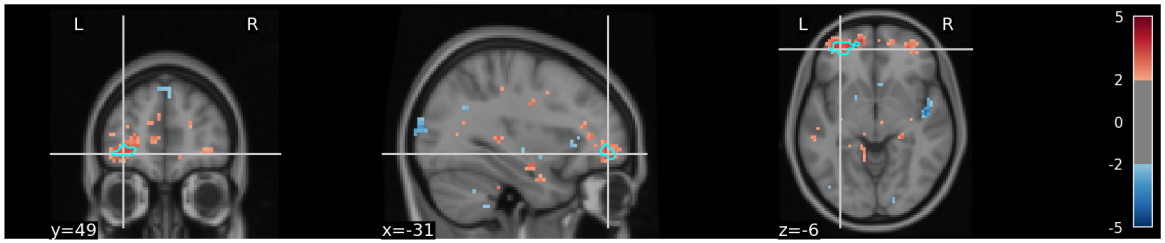
Content

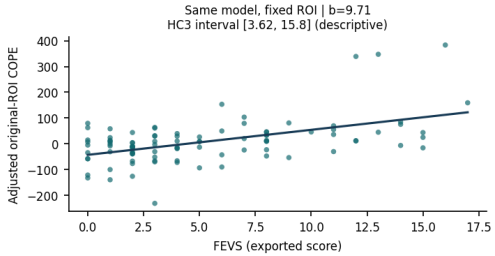
ecog | whole-brain cluster-FWE p=0.0666; corrected voxels=0





adi_ecog | whole-brain cluster-FWE p=0.0709; corrected voxels=0





Each brain row is beside its exact model-adjusted scatter. Cyan: original corrected cluster; gold: adjusted corrected cluster. The same original ROI is used for every scatter, including whole-brain null fits. Brain shading |t|>=2 is descriptive, not corrected significance.

Model	Cluster-FWE p, CDT 3.1	ROI slope / HC3 interval (descriptive)	Leave-one-out slope range / max Cook's D
ecog	0.0666	9.68 [3.76, 15.6]	[7.93, 10.3] / 0.214
adi_ecog	0.0709	9.71 [3.62, 15.8]	[7.92, 10.4] / 0.264

Conclusion

The original association is positive, but adding Ecog removes whole-brain corrected significance. Treat this as qualified evidence, not an adjustment-robust cortical vmPFC/OFC finding.

Anatomy caveat: original clusters are predominantly white-matter-labeled in a standard atlas; the jointly adjusted conditional cluster is approximately 64% white-matter-labeled. The conditional activation centroid shifts only 1.1 mm with joint adjustment. Neither fact validates a clean cortical vmPFC/OFC claim.

All 97 participants retained. Selected-ROI scatter slopes/HC3 intervals are descriptive only, not independent confirmation or a rescue of a null whole-brain result. No cases removed; leave-one-out checks do not exclude collective influence. Findings pass only at CDT 3.1; correction is per map, not across the exploratory model family. None of these focal models contains MSPSS.

3. Exploratory Activation

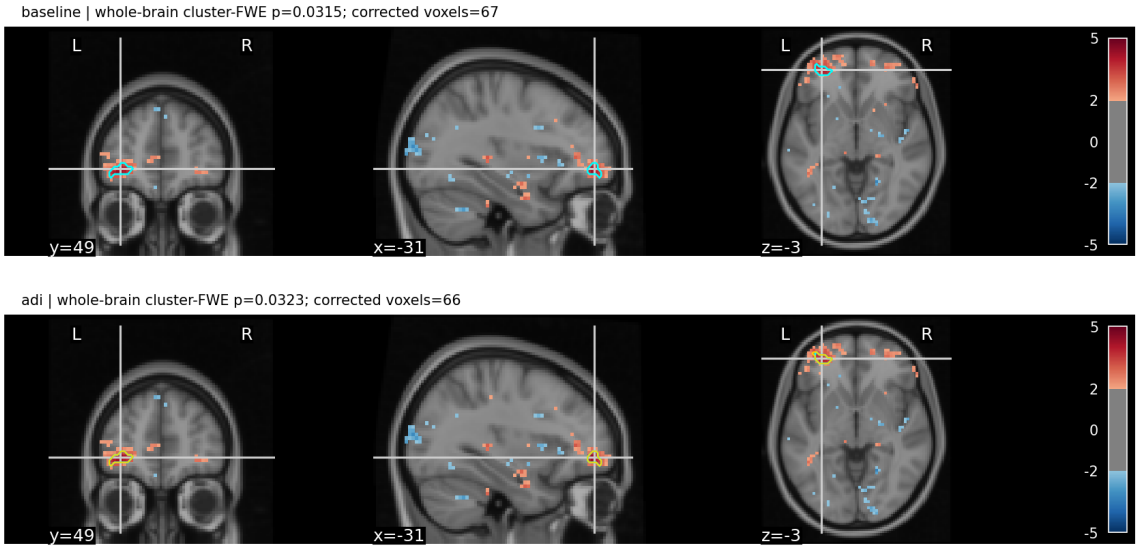
Does the FEVS activation effect remain when PM2.5 is modeled?

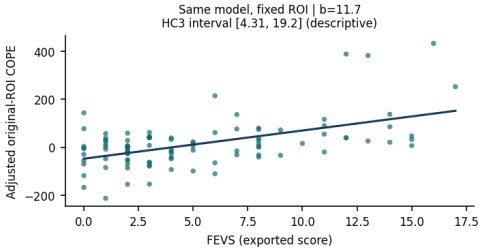
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

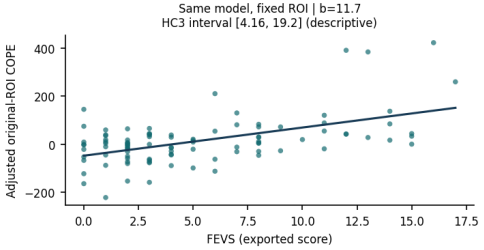
Context

Does the FEVS activation coefficient remain when PM2.5 and FEVS x PM2.5 are included? The coefficient is evaluated at mean PM2.5. Rows: original and +ADI.

Content







Each brain row is beside its exact model-adjusted scatter. Cyan: original corrected cluster; gold: adjusted corrected cluster. The same original ROI is used for every scatter, including whole-brain null fits. Brain shading |t|>=2 is descriptive, not corrected significance.

Model	Cluster-FWE p, CDT 3.1	ROI slope / HC3 interval (descriptive)	Leave-one-out slope range / max Cook's D
baseline	0.0315	11.7 [4.31, 19.2]	[9.7, 12.7] / 0.255
adi	0.0323	11.7 [4.16, 19.2]	[9.63, 12.6] / 0.322

Conclusion

The conditional FEVS coefficient retains a similar location, but correction is borderline and depends on adjustment. This is a main coefficient, not a significant winning-related FEVS x PM2.5 interaction.

Anatomy caveat: original clusters are predominantly white-matter-labeled in a standard atlas; the jointly adjusted conditional cluster is approximately 64% white-matter-labeled. The conditional activation centroid shifts only 1.1 mm with joint adjustment. Neither fact validates a clean cortical vmPFC/OFC claim.

All 97 participants retained. Selected-ROI scatter slopes/HC3 intervals are descriptive only, not independent confirmation or a rescue of a null whole-brain result. No cases removed; leave-one-out checks do not exclude collective influence. Findings pass only at CDT 3.1; correction is per map, not across the exploratory model family. None of these focal models contains MSPSS.

3. Exploratory Activation

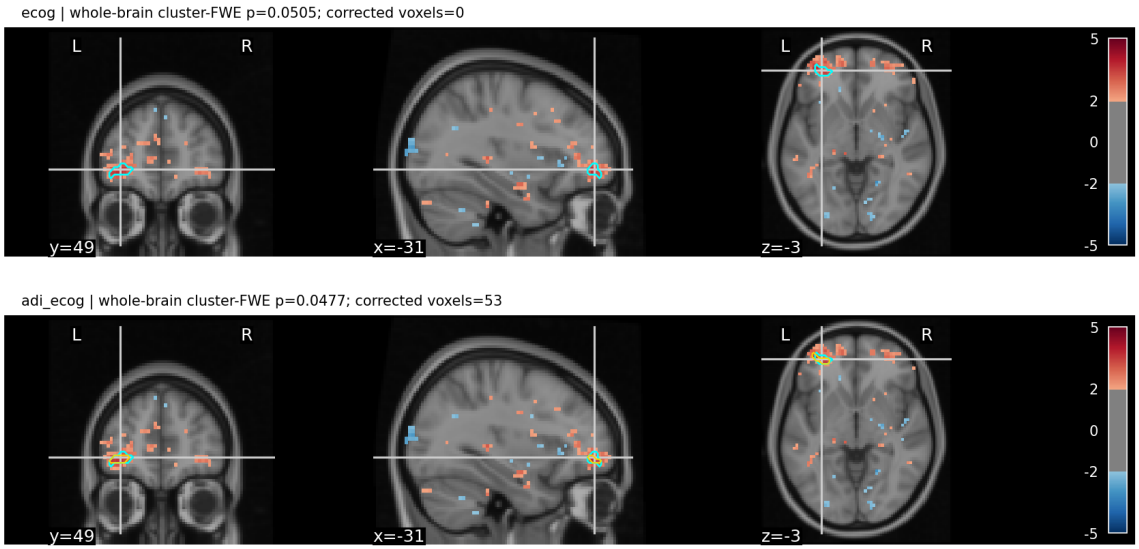
Does PM2.5-adjusted activation survive ADI/Ecog control?

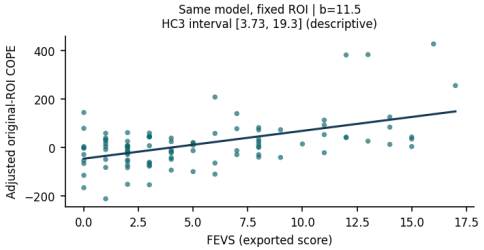
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

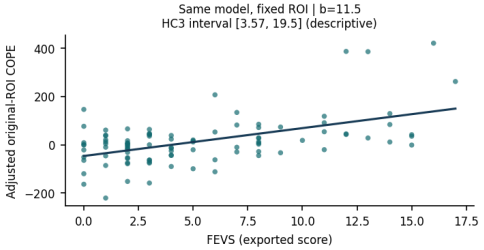
Context

Does the FEVS activation coefficient remain when PM2.5 and FEVS x PM2.5 are included? The coefficient is evaluated at mean PM2.5. Rows: +Ecog and +ADI/Ecog.

Content







Each brain row is beside its exact model-adjusted scatter. Cyan: original corrected cluster; gold: adjusted corrected cluster. The same original ROI is used for every scatter, including whole-brain null fits. Brain shading |t|>=2 is descriptive, not corrected significance.

Model	Cluster-FWE p, CDT 3.1	ROI slope / HC3 interval (descriptive)	Leave-one-out slope range / max Cook's D
ecog	0.0505	11.5 [3.73, 19.3]	[9.01, 12.4] / 0.359
adi_ecog	0.0477	11.5 [3.57, 19.5]	[9.03, 12.5] / 0.404

Conclusion

The conditional FEVS coefficient retains a similar location, but correction is borderline and depends on adjustment. This is a main coefficient, not a significant winning-related FEVS x PM2.5 interaction.

Anatomy caveat: original clusters are predominantly white-matter-labeled in a standard atlas; the jointly adjusted conditional cluster is approximately 64% white-matter-labeled. The conditional activation centroid shifts only 1.1 mm with joint adjustment. Neither fact validates a clean cortical vmPFC/OFC claim.

All 97 participants retained. Selected-ROI scatter slopes/HC3 intervals are descriptive only, not independent confirmation or a rescue of a null whole-brain result. No cases removed; leave-one-out checks do not exclude collective influence. Findings pass only at CDT 3.1; correction is per map, not across the exploratory model family. None of these focal models contains MSPSS.

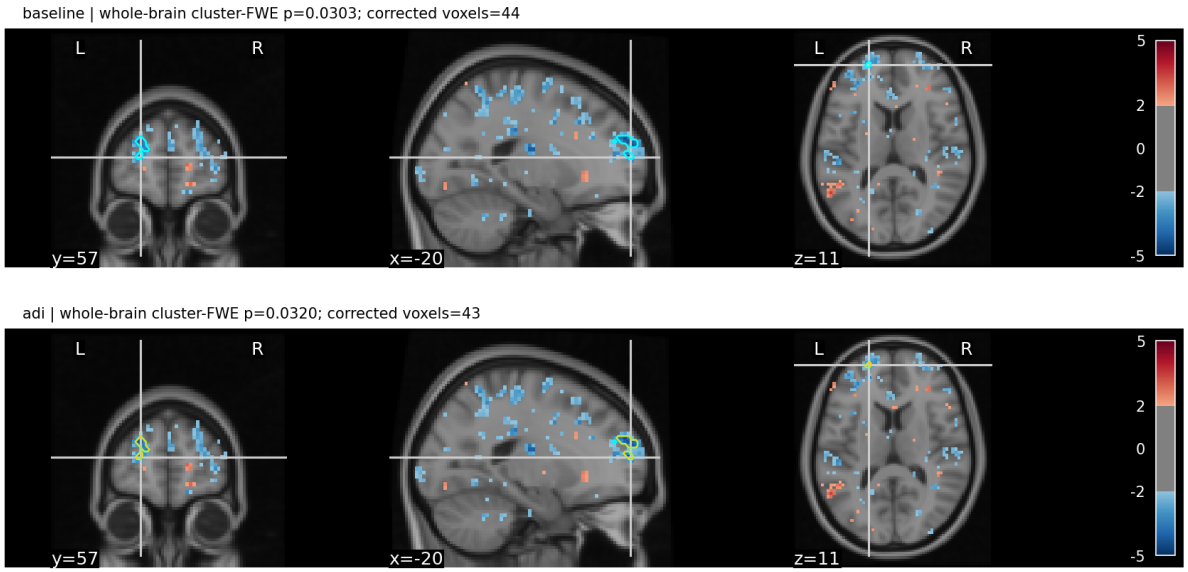
Does FEVS relate to socially differentiated VS connectivity?

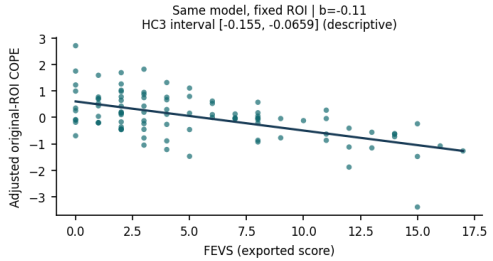
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

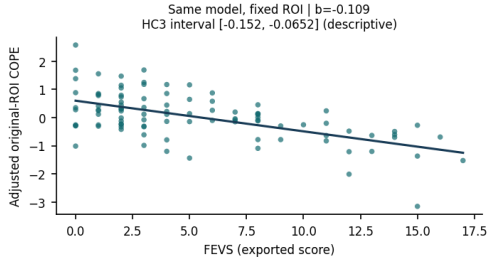
Context

Does financial vulnerability relate to VS coupling for (social win-loss) minus (nonsocial win-loss), at mean PM2.5? Rows: original and +ADI.

Content







Each brain row is beside its exact model-adjusted scatter. Cyan: original corrected cluster; gold: adjusted corrected cluster. The same original ROI is used for every scatter, including whole-brain null fits. Brain shading |t|>=2 is descriptive, not corrected significance.

Model	Cluster-FWE p, CDT 3.1	ROI slope / HC3 interval (descriptive)	Leave-one-out slope range / max Cook's D
baseline	0.0303	-0.11 [-0.155, -0.0659]	[-0.114, -0.0975] / 0.200
adi	0.0320	-0.109 [-0.152, -0.0652]	[-0.114, -0.0973] / 0.187

Conclusion

Higher FEVS relates to a lower social-by-valence connectivity contrast with a left frontal target. This is more consistent across ADI/Ecog adjustment than activation, but it is not evidence of globally weaker connectivity or a winning-only effect.

The left frontal target is not the VS seed. Joint adjustment retains 42 voxels, 32 overlapping the original 44; centroid shift=1.3 mm. Every adjusted peak remains inside the original cluster.

All 97 participants retained. Selected-ROI scatter slopes/HC3 intervals are descriptive only, not independent confirmation or a rescue of a null whole-brain result. No cases removed; leave-one-out checks do not exclude collective influence. Findings pass only at CDT 3.1; correction is per map, not across the exploratory model family. None of these focal models contains MSPSS.

Does the connectivity association persist with ADI/Ecog control?

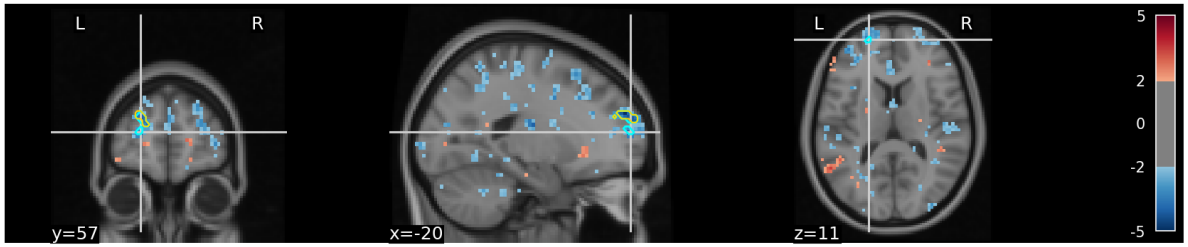
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

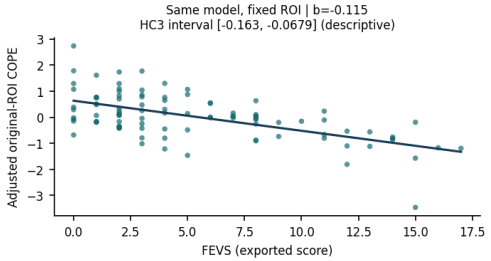
Context

Does financial vulnerability relate to VS coupling for (social win-loss) minus (nonsocial win-loss), at mean PM2.5? Rows: +Ecog and +ADI/Ecog.

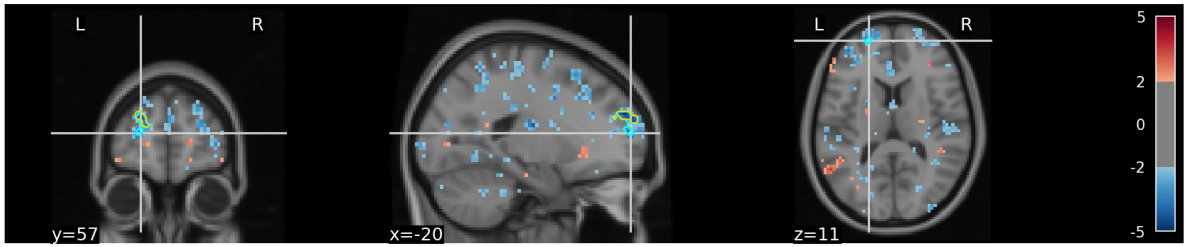
Content

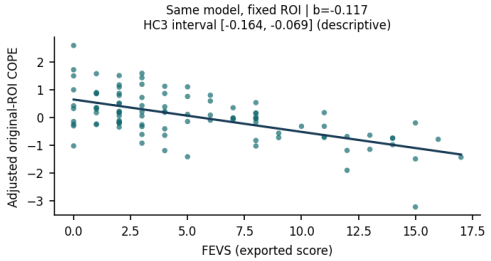
ecog | whole-brain cluster-FWE p=0.0349; corrected voxels=42





adi_ecog | whole-brain cluster-FWE p=0.0353; corrected voxels=42





Each brain row is beside its exact model-adjusted scatter. Cyan: original corrected cluster; gold: adjusted corrected cluster. The same original ROI is used for every scatter, including whole-brain null fits. Brain shading |t|>=2 is descriptive, not corrected significance.

Model	Cluster-FWE p, CDT 3.1	ROI slope / HC3 interval (descriptive)	Leave-one-out slope range / max Cook's D
ecog	0.0349	-0.115 [-0.163, -0.0679]	[-0.122, -0.103] / 0.186
adi_ecog	0.0353	-0.117 [-0.164, -0.069]	[-0.123, -0.105] / 0.173

Conclusion

Higher FEVS relates to a lower social-by-valence connectivity contrast with a left frontal target. This is more consistent across ADI/Ecog adjustment than activation, but it is not evidence of globally weaker connectivity or a winning-only effect.

The left frontal target is not the VS seed. Joint adjustment retains 42 voxels, 32 overlapping the original 44; centroid shift=1.3 mm. Every adjusted peak remains inside the original cluster.

All 97 participants retained. Selected-ROI scatter slopes/HC3 intervals are descriptive only, not independent confirmation or a rescue of a null whole-brain result. No cases removed; leave-one-out checks do not exclude collective influence. Findings pass only at CDT 3.1; correction is per map, not across the exploratory model family. None of these focal models contains MSPSS.

Appendix: Design Validation

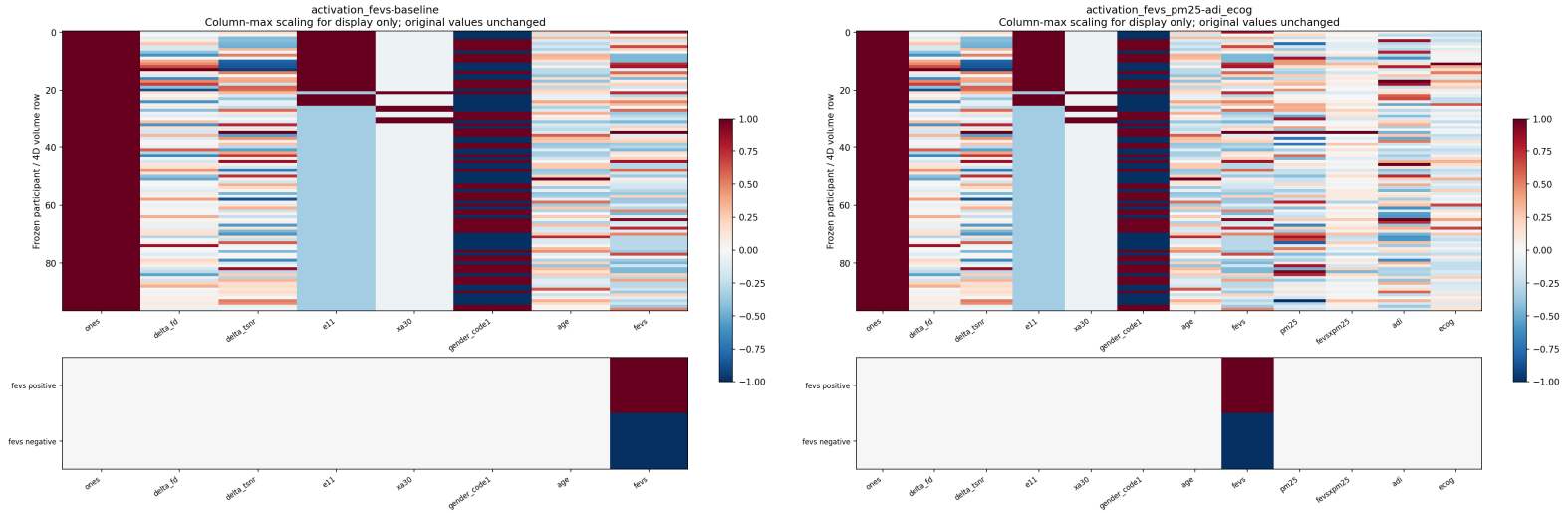
Did the group designs test the intended effects?

PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Check the actual contrast weights, rank and image-volume ordering rather than inferring the test from the presence of a column of ones.

Content



Left: standalone FEVS. Right: FEVS/PM2.5 with additive ADI and Ecog. These published heatmaps read actual executed .mat/.con files, not newly chosen designs, and are not native FEAT screenshots. Display-only column scaling; participant identifiers omitted. The separate complete audit verified image/design ordering.

Executed model	Rank / residual df	Condition number (scaled)	Intercept tested?
activation_fevs-baseline	8 / 89	2.14	No
activation_fevs_pm25-baseline	10 / 87	2.23	No
activation_fevs_pm25-adi_ecog	12 / 85	2.39	No
task-mean-intercept	7 / 90	2.14	Yes

Check	Verdict
Column of ones	Included, but NOT tested in the plotted FEVS +/- contrasts. An intercept must have its own nonzero contrast weight.
FEVS coefficient versus interaction	Standalone plot tests FEVS. PM2.5-containing figures test FEVS at mean PM2.5, not FEVS x PM2.5.
Preserved	Frozen N=97, settled exclusions, all completed imaging results, and original PDF versions.

Conclusion

The current coherent description is provisional financial-vulnerability associations with socially differentiated brain responses. Activation is qualified; the VS-connectivity sensitivity result is more consistent. The environmental-support behavioral link awaits corrected MSPSS scoring.

Were the intended win and loss outcomes modeled?

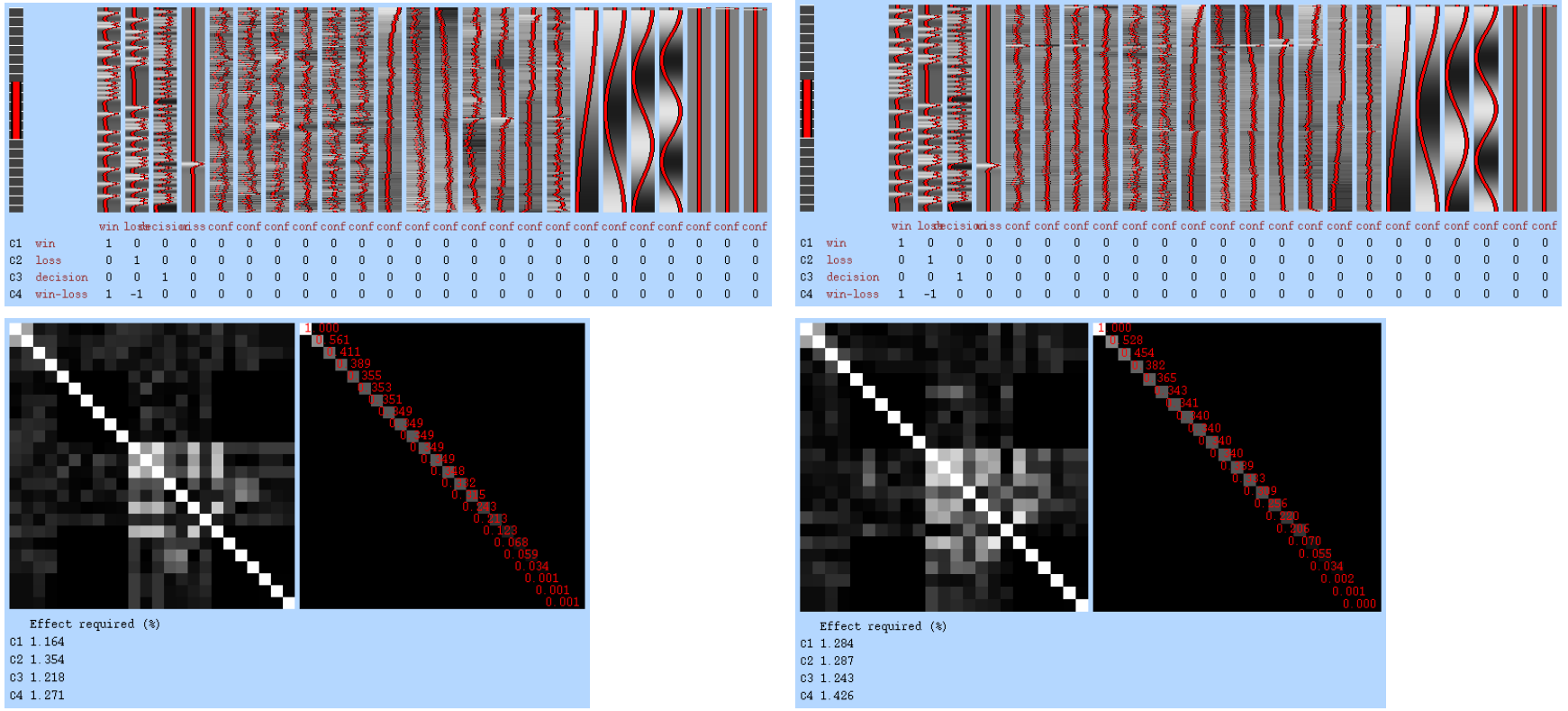
PRELIMINARY | Frozen N=97 | Exploratory, not preregistered

Context

Inspect unmodified native FEAT screenshots from the first frozen participant's two activation L1s, not custom group-design heatmaps.

Content

Social Doors	Nonsocial Doors



Top: native design.png (design and contrast display). Bottom: native design_cov.png. Both tasks' actual design.fsf/.mat/.con were checked for the expected win, loss and win-minus-loss contrast weights before these PNGs were copied unchanged. These are representative design views, not evidence that every participant has identical event timing.

Conclusion

The expected outcome contrasts are present, and all requested evidence artifacts passed the saved-input checks. N=97, previous analyses and exclusions remain unchanged. The remaining measurement issue is scoring verification and the planned comparison of MSPSS administrations, not a new imaging exclusion.